Example 3.1 (Introduction): Let us look at a simple LP problem:

$$
\begin{aligned}
\text{Maximize} \quad & z = f(x_1, x_2) = x_1 + x_2 \\
\text{subject to} \quad & -x_1 + x_2 \le 1 \\
& x_1 \le 3 \\
& x_2 \le 2 \\
& x_1, x_2 \ge 0
\end{aligned}
\qquad (3.1)
$$

We change this problem to <u>canonical (equational) form</u>. For this, we only need to introduce slack variables, which here we call $\underbrace{x_3, x_4, x_5}_{\text{slack variables}}$. We get:

Cost function, objective function

Slack variables

$$
\begin{array}{lll}
\text{Maximize} & z = x_1 + x_2 & (*) \\
\text{subject to} & (a) \quad -x_1 + x_2 + \boxed{x_3} = 1 & \\
& (b) \quad x_1 \boxed{+ x_4} = 3 & \\
& (c) \quad x_2 \boxed{+ x_5} = 2 & \\
& x_1, x_2, x_3, x_4, x_5 \geq 0 &
\end{array}
\tag{3.2}
$$

Recall: A feasible solution is a vector

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

that satisfies all these constraints

In matrix form,

$$\text{Max.} \quad z = \mathbf{c}^T \mathbf{x} \qquad \text{subject to} \qquad A\mathbf{x} = \mathbf{b}, \qquad \mathbf{x} \geq 0$$

where $\mathbf{c}^T = [\overbrace{1 \;\; 1 \;\; 0 \;\; 0 \;\; 0}^{\text{See cost function } (*)}]$

$$A = \begin{bmatrix} -1 & 1 & \boxed{1 & 0 & 0} \\ 1 & 0 & \boxed{0 & 1 & 0} \\ 0 & 1 & \boxed{0 & 0 & 1} \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

A_B this part of A comes from the slack variables

We note that the matrix A has rank 3 (= 3 linearly independent rows.)

$B = \{3, 4, 5\}$

We keep in mind that all $x_i \geq 0$, and write (3.2) in form of a *simplex tableau*, taking the slack variables to the left:

from (a): $$x_3 = 1 + x_1 - x_2 \qquad (3.3a)$$

from (b): $$x_4 = 3 - x_1 \qquad (3.3b)$$

from (c): $$x_5 = 2 \qquad - x_2 \qquad (3.3c)$$

$$z = \qquad x_1 + x_2 \qquad (3.3d)$$

when x_1 or x_2 get larger, then z also gets larger

when $x_1 = 0, x_2 = 0$

We can see that when we set the variables on the right (x_1 and x_2) to zero, then $\boxed{x_3 = 1, \quad x_4 = 3, \quad x_5 = 2.}$ We have found <u>one</u> feasible solution of special form: $(x_1, x_2, x_3, x_4, x_5) = (0, 0, 1, 3, 2)$ (*).

We write the {indices of the 3 variables on the left} as a set $B = \{3, 4, 5\}$ and extract the corresponding columns from the matrix A as

$$A_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We call

- solution (*) [where $x_i = 0$ for $i \notin B$] a *basic feasible solution*,
- the variables x_i with $i \in B$ which are the variables on the left side in the tableau [here x_3, x_4, x_5] *basic variables.*
- the variables x_i with $i \notin B$ which are the variables on the right side in the tableau [here x_1, x_2] *nonbasic variables.*
- the set $B = \{3, 4, 5\}$ a *basis.*

Recall that we want to maximize z. To do so, we will modify tableau (3.3) in several steps. If we look at the objective function (3.3d), we notice that increasing x_1 and/or x_2 to a positive number will increase the value of z.

We choose to increase x_2 (and so keep $x_1 = 0$).

(3.3c) $x_5 = 2 - x_2$. Because we need $x_5 \geq 0$ therefore $\boxed{x_2 \leq 2}$

(3.3a) $x_3 = 1 + x_1 - x_2 = 1 - x_2$ Because we need $x_3 \geq 0$ therefore $\boxed{x_2 \leq 1}$

when $x_1 = 0$

Can increas x_2 to $x_2 = 1$ only

By equation (3.3c), we can increase x_2 only up to $x_2 = 2$ (because it is required that $x_5 \geq 0$).

By equation (3.3a), we can increase x_2 only up to $x_2 = 1$ (because it is required that $x_3 \geq 0$ (recall that here $x_2 = 0$!)) (3.3a) is the equation to consider.

When we increase x_2, equation (3.3b) is unchanged.

Thus , we increase x_2 from 0 to $x_2 = 1$, as equation (3.3a) allows.

- From equation (3.3a) when $x_2 = 1$ we obtain $x_3 = 0$, which becomes a new nonbasic variable. On the other hand, solving (3.3a) for x_2, this will become a new basic variable: We make x_2 a basic variable

(3.3a): $x_3 = 1 + x_1 - x_2$

Now solve for x_2: $x_2 = 1 + x_1 - x_3$ (*) (3.4a)

replace all x_2 on the right sides of (3.3)

- equation (3.3b) remains unchanged
- equation (3.3c) becomes $x_5 = 2 - (1 + x_1 - x_3) = 1 - x_1 + x_3$ (3.4a)
- the objective function becomes $z = x_1 + (1 + x_1 - x_3) = 1 + 2x_1 - x_3$ (3.4d)

replace all x_2 on the right

$B = \{2, 4, 5\}$

We obtain a new simplex tableau

$$x_2 = 1 + x_1 - x_3 \qquad (3.4a)$$

$$x_4 = 3 - x_1 \qquad (3.4b)$$

$$x_5 = 1 - x_1 + x_3 \qquad (3.4c)$$

basic variables

$$z = 1 + 2x_1 - x_3 \qquad (3.4d)$$

When x_1 gets larger, then z gets larger

But when x_3 gets larger, then z gets smaller !!

$\Downarrow$

So in the next step, we increase x_1.

We have found a <u>new</u> basic feasible solution:

$$(x_1, x_2, x_3, x_4, x_5) = (0, 1, 0, 3, 1)$$

x_1 x_3

$x_1 = 0$ $x_3 = 0$

- with basis $B = \{2, 4, 5\}$
- basic variables x_2, x_4, x_5
- nonbasic variables x_1, x_3
- value of $\boxed{z = 1}$ [as $x_1 = 0, x_3 = 0$]

(3.4c) $x_5 = 1 - x_1 + x_3 = 1 - x_1$ (when $x_3 = 0$)

Because we need $x_5 \geq 5$, then $x_1 \leq 1$ is required

(3.4b): $x_4 = 3 - x_1$. Because we need $x_4 \geq 0$, so $x_1 \leq 3$ is required

(3.4c) limits to $x_1 \leq 1$

Looking at (3.4d) we can only increase x_1 (we can't increase x_3 !!).

Equation (3.4a), allows increase of x_1 to <u>any</u> positive value.

- By equation (3.4b), we can increase x_1 only up to $x_1 = 3$
- By equation (3.4c), we can increase x_1 only up to $x_1 = 1$

Thus , we increase x_1 from 0 to $x_1 = 1$, as equation (3.4c) allows.

- From equation (3.4c) when $x_1 = 1$ we obtain $x_5 = 0$, which becomes a new nonbasic variable. On the other hand, solving (3.4c) for x_1, this will become a new basic variable:

(3.4c): $x_5 = 1 - x_1 + x_3$

Solve for x_1:

$$x_1 = 1 + x_3 - x_5 \qquad (*) \qquad (3.5c)$$

replace all x_1 on the right side

- equation (3.4b) becomes $x_4 = 3 - (1 + x_3 - x_5) = 2 - x_3 + x_5$ (3.5b)
- equation (3.4a) becomes $x_2 = 1 + (1 + x_3 - x_5 - x_3) = 2 - x_5$ (3.5a)
- the objective function becomes

$$z = 1 + 2\underbrace{(1 + x_3 - x_5)}_{\text{replace } x_1} - x_3 = 3 + x_3 - 2x_5$$

$B = \{1, 2, 4\}$

We obtain a new simplex tableau

$$x_1 = 1 + x_3 - x_5 \qquad (3.5a)$$

$$x_2 = 2 \qquad - x_5 \qquad (3.5b)$$

$$x_4 = 2 - x_3 + x_5 \qquad (3.5c)$$

$$z = 3 + x_3 - 2x_5 \qquad (3.5d)$$

↑ when x_3 increases, then z increases

We have found a <u>new</u> basic feasible solution:

$$(x_1, x_2, \underset{=}{x_3}^{=0}, x_4, \underset{=}{x_5}^{=0}) = (1, 2, 0, 2, 0)$$

- with basis $B = \{1, 2, 4\}$
- basic variables x_1, x_2, x_4
- nonbasic variables x_3, x_5
- value of $\boxed{z = 3}$ (as $x_3 = x_5 = 0$)

(3.5c) $x_4 = 2 - x_3 + x_2 = 2 - x_3$ (when $x_5 = 0$)

it is required that $x_4 \geq 0$ $\Rightarrow x_3 \leq 2$

Looking at (3.5d) we can only increase x_3 (we can't increase x_5 !!).

- Equation (3.5a), allows to increase of x_3 by as much as we wish.

  Equation (3.5b) has no x_3.

- By equation (3.5c), we can increase x_3 only up to $x_3 = 2$.

  Then $x_4 = 0$ and x_4 becomes a nonbasic variable ("inactive"), while x_3 becomes a basic variable ("active"). Solve (3.5c) for x_3:

(3.5c): $x_4 = 2 - x_3 + x_2 \Rightarrow$

$$x_3 = 2 - x_4 + x_5 \qquad (3.6c)$$

- (3.5a) changes to: $x_1 = 1 + (2 - x_4 + x_5) - x_5 = 3 - x_4$ (3.6a)
- (3.5b) does not change. (3.6b)
- the objective (3.5d) function becomes

$$z = 3 + \underbrace{(2 - x_4 + x_5)}_{x_3} - 2x_5 = 5 - x_4 - x_5 \qquad (3.6d)$$

We obtain a new simplex tableau:

$B = \{1, 2, 3\}$

$$x_1 = 3 - x_4 \qquad (3.6a)$$

$$x_2 = 2 \qquad - x_5 \qquad (3.6b)$$

$$x_3 = 2 - x_4 + x_5 \qquad (3.6c)$$

$$z = 5 - x_4 - x_5 \qquad (3.6d)$$

And we have found a <u>new</u> basic feasible solution:

$$(x_1, x_2, x_3, x_4, x_5) = (3, 2, 2, 0, 0)$$

- with basis $B = \{1, 2, 3\}$
- basic variables x_1, x_2, x_3, nonbasic variables x_4, x_5
- value of $\boxed{z = 5.}$ (when $x_4 = x_5 = 0$)

From (3.6d) we notice that z can no longer be increased. We stop.

Looking back at problem (3.1), we observe that our solution

$$\boxed{x_1 = 3, \ x_2 = 2 \ \text{ and } \ z = 5}$$

is <u>optimal.</u>

optimal feasible variables

max value of cost function

Let us see <u>graphically</u> what the simplex algorithm does.

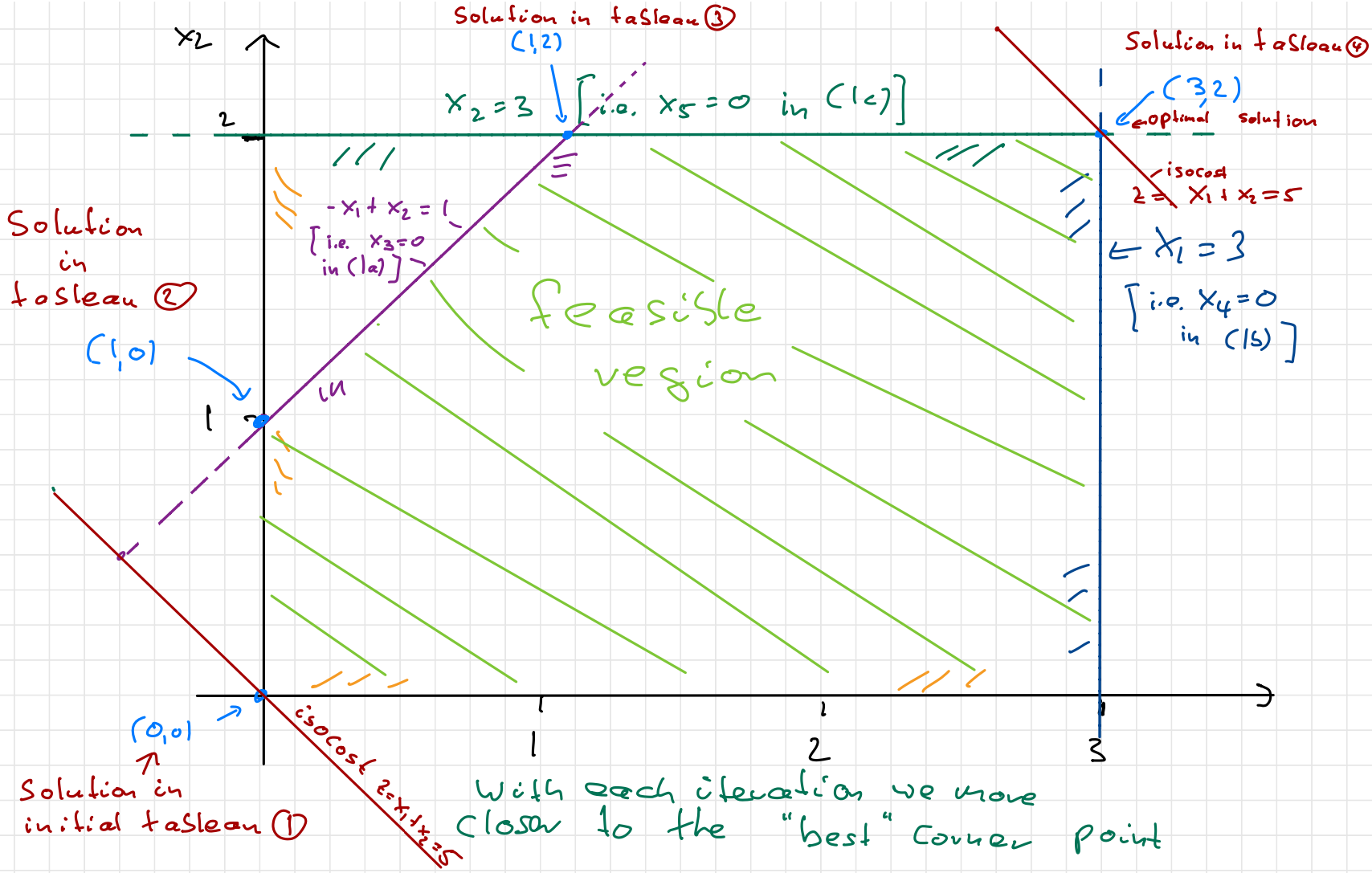
The <u>original</u> LP-problem (3.1) was:

$$\begin{cases} \text{Maximize} & z = x_1 + x_2 \\ \text{Subject to:} & -x_1 + x_2 \le 1 \quad (a) \\ & x_1 \le 3 \quad (b) \\ & x_2 \le 2 \quad (c) \\ & x_1, x_2 \ge 0 \quad (d) \end{cases}$$

After introducing slack variables x_3, x_4, x_5:

$$\begin{cases} \text{Maximize} & z = x_1 + x_2 \\ \text{Subject to:} & -x_1 + x_2 + x_3 = 1 \quad (1a) \\ & x_1 + x_4 = 3 \quad (1b) \\ & x_2 + x_5 = 2 \quad (1c) \\ & x_1, x_2, x_3, x_4, x_5 \ge 0 \end{cases}$$

Sketch the feasible region:


Solution in tableau 3
(1,2)
$x_2 = 3$ [i.e. $x_5 = 0$ in (1c)]
Solution in tableau 4
(3,2)
optimal solution
isocost
$z = x_1 + x_2 = 5$
$x_1 = 3$
[i.e. $x_4 = 0$ in (1b)]
x_2
2
$-x_1 + x_2 = 1$
[i.e. $x_3 = 0$ in (1a)]
Solution in tableau 2
(1,0)
1
feasible region
1
2
3
(0,0)
isocost $z = x_1 + x_2 = 5$
Solution in initial tableau 1
With each iteration we move closer to the "best" corner point

Exercise: Maximize $z = x_1 + 2x_2$

Subject to:

$$\begin{cases} x_1 + 4x_2 \leq 20 \\ x_1 + x_2 \leq 8 \\ 5x_1 + x_2 \leq 32 \end{cases}$$

$$x_1, x_2 \geq 0$$

Change to canonical form:

Max. $z = x_1 + 2x_2$

Slack variables

Subject to:

$$\begin{cases} x_1 + 4x_2 + x_3 = 20 \\ x_1 + x_2 + x_4 = 8 \\ 5x_1 + x_2 + x_5 = 32 \end{cases}$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

(1) Write a simplex tableau starting with the slack variables as basic variables.

$\boxed{\text{Tableau (1)}}$

$$x_3 = 20 - x_1 - 4x_2 \qquad (1a)$$
$$x_4 = 8 - x_1 - x_2 \qquad (1b)$$
$$x_5 = 32 - 5x_1 - x_2 \qquad (1c)$$
$$\overline{}$$
$$z = x_1 + 2x_2 \qquad (1d)$$

A basic solution [here $x_1 = x_2 = 0$] is

$$(x_1, x_2, x_3, x_4, x_5) = (0, 0, 20, 8, 32)$$

A basis is $B = \{3, 4, 5\}$

- basic variables: x_3, x_4, x_5
- nonbasic variables: x_1, x_2
- value of cost: $\boxed{z = 0 + 0 = 0}$

Looking at (1d): we can increase z by increasing x_1 or x_2.

We increase x_2. What is the max possible increase so that $x_3, x_4, x_5 \geq 0$?

By (1a): $0 = 20 - 4x_2 \Rightarrow 4x_2 = 20 \rightarrow \boxed{x_2 = 5}$

$(x_1 = 0)$

By (1b): $0 = 8 - x_2 \Rightarrow \boxed{x_2 = 8}$

By (1c): $32 - x_2 = 0 \Rightarrow \boxed{x_2 = 50}$

$x_2 = 5$ is smallest, so x_2 is limited by equation (1a). So we make x_2 a basic variable in (1a)

In (1a), make x_2 a slack variable $\Rightarrow x_3$ becomes nonbasic

By (1a): $x_3 = 20 - x_1 - 4x_2$

$\Rightarrow 4x_2 = 20 - x_1 - x_3$

$$\boxed{x_2 = 5 - \frac{1}{4}x_1 - \frac{1}{4}x_3}$$ New (2a)

Substitute (2a) into (1b), (1c), and (1d)

(1b): $x_4 = 8 - x_1 - \left[5 - \frac{1}{4}x_1 - \frac{1}{4}x_3\right] = 3 - \frac{3}{4}x_1 + \frac{1}{4}x_2$ New (2b)

(1c): $x_5 = 32 - 5x_1 - \left[5 - \frac{1}{4}x_1 - \frac{1}{4}x_3\right] = 27 - \frac{19}{4}x_1 + \frac{1}{4}x_3$ New (2c)

(1d) $z = x_1 + 2\left[5 - \frac{1}{4}x_1 - \frac{1}{4}x_3\right] = 10 + \frac{1}{2}x_1 - \frac{1}{2}x_3$ (2d)

We obtain a new tableau:

$\boxed{\text{Tableau ②}}$

$$x_2 = 5 - \frac{1}{4}x_1 - \frac{1}{4}x_3 \qquad (2a)$$

$$x_4 = 3 - \frac{3}{4}x_1 + \frac{1}{4}x_3 \qquad (2b)$$

$$x_5 = 27 - \frac{19}{4}x_1 + \frac{1}{4}x_3 \qquad (2c)$$

$$\hline$$

$$z = \underline{\underline{10}} + \frac{1}{2}x_1 - \frac{1}{2}x_3 \qquad (2d)$$

The basic solution is $(x_1, x_2, x_3, x_4, x_5) = (0, 5, 0, 3, 27)$

- with basis $B = \{2, 4, 5\}$
- basic variables x_2, x_4, x_5
- nonbasic variables x_1, x_3
- value of cost function $\boxed{z = 10}$

③ Looking at the cost function (2d), we can only increase x_1. What is the max possible increase?

In (2a): $5 - \frac{1}{4}x_1 = 0 \Rightarrow \frac{1}{4}x_1 = 5 \Rightarrow \boxed{x_1 = 20}$

In (2b): $3 - \frac{3}{4}x_1 = 0 \Rightarrow \frac{3}{4}x_1 = 3 \Rightarrow \boxed{x_1 = 4}$

In (2c): $27 - \frac{19}{4}x_1 = 0 \Rightarrow \frac{19}{4}x_1 = 27 \Rightarrow \boxed{x_2 = \frac{108}{19} = 5\frac{13}{19}}$

↳ The max possible value is $x_1 = 4$, limited by equation (2b).

So we make x_1 a basic variable in (2b)

(2b): $x_4 = 3 - \frac{3}{4}x_1 + \frac{1}{4}x_3 \Rightarrow \frac{3}{4}x_1 = 3 + \frac{1}{4}x_3 - x_4$

$$\Rightarrow \boxed{x_1 = 4 + \frac{1}{3}x_3 - \frac{4}{3}x_4} \qquad (3b)$$

Replace x_1 in the remaining equations:

(2a): $x_2 = 5 - \frac{1}{4}\left[4 + \frac{1}{3}x_3 - \frac{4}{3}x_4\right] - \frac{1}{4}x_3 = 4 - \frac{1}{3}x_3 + \frac{1}{3}x_4$ (3a)

(2c): $x_5 = 27 - \frac{19}{4}\left[4 + \frac{1}{3}x_3 - \frac{4}{3}x_4\right] + \frac{1}{4}x_3 = 8 - \frac{4}{3}x_3 + \frac{19}{4}x_4$ (3c)

(2d): $z = 10 + \frac{1}{2}\left[4 + \frac{1}{3}x_3 - \frac{4}{3}x_4\right] - \frac{1}{2}x_3 = 12 - \frac{1}{3}x_3 - \frac{2}{3}x_4$ (3d)

We obtain the simplex tableau:

$$
\begin{aligned}
x_1 &= 4 + \tfrac{1}{3}x_3 - \tfrac{4}{3}x_4 && (3b)\\
x_2 &= 4 - \tfrac{1}{3}x_3 + \tfrac{1}{3}x_4 && (3a)\\
x_5 &= 8 - \tfrac{4}{3}x_3 + \tfrac{19}{4}x_4 && (3c)\\
\hline
z &= 12 - \tfrac{1}{3}x_3 - \tfrac{2}{3}x_4 && (3d)
\end{aligned}
$$

This basic feasible solution is

$$(x_1, x_2, x_3, x_4, x_5) = (4, 4, 0, 0, 8)$$

with basis $B = \{1, 2, 5\}$,

basic variables x_1, x_2, x_5

nonbasic variables x_3, x_4

and value of cost $\boxed{z = 12}$

Looking at (3d) we see that we can't increase the value of z further (as $x_3 \geq 0$, $x_4 \geq 0$). So we stop.

<u>Answer</u>: At $x_1 = 4$, $x_2 = 4$, the max value of cost function is $\underline{\underline{z = 12}}$